

UNIT - IISHM & EMT## MAXWELL'S EQUATIONS -DIFFERENTIAL FORM

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{D} = \rho \quad \left\{ \begin{array}{l} \text{based on } \rightarrow \text{Gauss' Law of Electrostatics} \end{array} \right\}$$

$$\textcircled{2} \quad \vec{\nabla} \cdot \vec{B} = 0 \quad \left\{ \text{Gauss' Law of Magnetostatics} \right\}$$

$$\textcircled{3} \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \left\{ \text{Faraday's Law of EMMI} \right\}$$

$$\textcircled{4} \quad \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \left\{ \text{Modified Ampere's Circuital Law} \right\}$$

INTEGRAL FORM

$$\textcircled{1} \quad \oint_S \vec{E} \cdot d\vec{S} = \int_V \rho \, dv \quad \left(= \frac{q}{\epsilon_0} \right) \quad \left/ \quad \oint_S \vec{D} \cdot d\vec{S} = \int_V \rho \, dv \right.$$

$$\textcircled{2} \quad \oint_S \vec{B} \cdot d\vec{S} = 0$$

$$\textcircled{3} \quad \oint_C \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$$

$$\oint_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \oint_S \vec{B} \cdot d\vec{S}$$

$$\textcircled{4} \quad \oint_C \vec{H} \cdot d\vec{l} = \oint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S}$$

(I) 1st MAXWELL EQUATION — $\nabla \cdot \vec{D} = \rho$

- based on - Gauss' Law of electrostatics

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

$$\Rightarrow \oint_S \epsilon_0 \vec{E} \cdot d\vec{S} = \int_V \rho dV \quad \left\{ \because q = \int_V \rho dV \right\}$$

$$\Rightarrow \oint_S \vec{D} \cdot d\vec{S} = \int_V \rho dV$$

Volume charge density

Applying Gauss Divergence theorem — (in LHS) $(S \rightarrow V)$ $\left\{ \because \vec{D} = \epsilon_0 \vec{E} \right\}$
Displacement Density

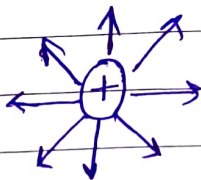
$$\Rightarrow \int_V \nabla \cdot \vec{D} dV = \int_V \rho dV$$

$$\Rightarrow \boxed{\nabla \cdot \vec{D} = \rho} \text{ ————— Maxwell eqⁿ \rightarrow 1}$$

- PHYSICAL SIGNIFICANCE - divergence of displacement charge density is always equal to volume charge density.

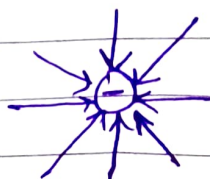
- charge acts as a source or sink for electric field lines

✓ positive charge field lines



Diverge away from charge

✓ negative charge field lines



Converge into the charge

- relates electric flux w/ charge

(II) IInd MAXWELL EQUATION — $\vec{\nabla} \cdot \vec{B} = 0$

- based on Gauss's law of magnetostatics

$$\boxed{\phi_B = \oint_S \vec{B} \cdot d\vec{S} = 0} \quad (1) \quad \text{(only valid for closed surface)}$$

- Applying Gauss' Divergence theorem —

$$\Rightarrow \int_V (\vec{\nabla} \cdot \vec{B}) dV = 0$$

$$\left[\begin{array}{l} \because B = \frac{\phi_B}{\text{area}} \rightarrow \text{mag. flux} \\ \downarrow \\ \text{mag. field} \end{array} \right]$$

$$\Rightarrow \boxed{\vec{\nabla} \cdot \vec{B} = 0} \quad \text{—— Maxwell eqⁿ → 2}$$

- PHYSICAL SIGNIFICANCE -

- Magnetic flux through closed surface is equal to zero.
- Reason why ^{isolated} magnetic monopoles don't exist

(III) IIIrd MAXWELL EQUATION — $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

- based on Faraday's law of EMMI (Electromagnetic Mutual Induction)
- (emf depends on negative rate change of ϕ_B)

~~EMF~~ $\text{emf} = -\frac{d(\phi_B)}{dt} \quad \text{--- (1)}$

$\text{emf} = \oint \vec{E} \cdot d\vec{L} \quad \text{--- (2)}$

equating eq's — (1) & — (2) —

$$\begin{aligned} \Rightarrow \oint \vec{E} \cdot d\vec{L} &= -\frac{d\phi_B}{dt} \\ \Rightarrow \oint \vec{E} \cdot d\vec{L} &= -\frac{d}{dt} \left(\oint \vec{B} \cdot d\vec{S} \right) \end{aligned} \quad \left\{ \begin{array}{l} \because B = \frac{d\phi_B}{A} \\ \phi_B = B \cdot A \\ = B \cdot dS \end{array} \right\}$$

$$\Rightarrow \oint_L \vec{E} \cdot d\vec{L} = - \oint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

Applying Stoke's Theorem — (in LHS) $(L \rightarrow S)$ partial diffⁿ, $\because$ because B-variable qty. $\rightarrow$ depends on more than

$$\Rightarrow \oint_S (\nabla \times \vec{E}) \cdot d\vec{S} = - \oint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\Rightarrow \boxed{\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}} \quad \text{--- Maxwell eqⁿ $\rightarrow$ 3}$$

- PHYSICAL SIGNIFICANCE -

- time variation of magnetic field generates electric field.
- negative sign justifies existence of Lenz law
- (- curl of $\vec{E}$ always equal to -ve rate change of magnetic flux density)

(IV) IVth MAXWELL EQUATION — $\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

— based on modified form of Ampere's circuital law (& continuity eqⁿ)

(line integral of magnetic field density ($\vec{B}$) along an imaginary closed path is equal to the product of current enclosed by path & permeability of medium)

$$\Rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$\Rightarrow \left[\oint \vec{H} \cdot d\vec{l} = I \right] \text{ — (1) } \left\{ \because \vec{B} = \mu_0 \vec{H} \right\}$$

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = \oint \vec{J} \cdot d\vec{S} \quad \left\{ \because I = \oint \vec{J} \cdot d\vec{S} \right\}$$

Applying Stoke's curl theorem — (in LHS)

$$\Rightarrow \oint (\vec{\nabla} \times \vec{H}) \cdot d\vec{S} = \oint \vec{J} \cdot d\vec{S}$$

$$\Rightarrow \boxed{\vec{\nabla} \times \vec{H} = \vec{J}} \text{ — (a)}$$

to check validity of eqⁿ — (a) by eqⁿ of continuity of i.e. — $\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$

for — (a) to be valid, $\vec{\nabla} \cdot \vec{J} \neq 0$

Taking divergence of — (a) —

$$\Rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J}$$

from vector identity $\rightarrow \text{div}(\text{curl } \vec{A}) = 0$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} = 0$$

$$\oint \vec{H} \cdot d\vec{l} = \oint \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) d\vec{s}$$

But, from eqⁿ of continuity, $\vec{\nabla} \cdot \vec{J} \neq 0$
Hence, we can say eqⁿ — (a) is invalid.

However, if conduction current added along w/
introduced concept of displacement current, eqⁿ — (a) will be valid

(So, Maxwell Modified Ampere's Circital Law)

⇒ Instead of only $\rightarrow \vec{\nabla} \times \vec{H} = \vec{J}$

⇒ $\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}_d$ (b) $\{ \vec{J}_d \rightarrow \text{displacement current density} \}$

Again, checking validity of eqⁿ — (b),

Taking divergence of — (b) —

$$\Rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}_d$$

$$\Rightarrow 0 = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}_d$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} = -\vec{\nabla} \cdot \vec{J}_d \quad \text{--- (c)}$$

Also, from eqⁿ of continuity, $\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$

So, $\vec{\nabla} \cdot \vec{J} \neq 0$ for eqⁿ — (b), Hence from eqⁿ of continuity, it is a valid eqⁿ.

Now, ~~$\vec{\nabla} \times \vec{E} = -\vec{\nabla} \times \vec{A}$~~
equating eqⁿ — (c) w/ eqⁿ of continuity,

$$\Rightarrow -\vec{\nabla} \cdot \vec{J}_d = -\frac{\partial \rho}{\partial t}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J}_d = \frac{\partial \rho}{\partial t}$$

from Maxwell's first eqⁿ — $\vec{\nabla} \cdot \vec{D} = \rho$

$$\Rightarrow \vec{\nabla} \cdot \vec{J}_d = \frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} = \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\Rightarrow \vec{\nabla} \cdot \left(\vec{J} - \frac{\partial \vec{D}}{\partial t} \right) = 0 \Rightarrow \boxed{\vec{J} = \frac{\partial \vec{D}}{\partial t}} \quad \text{--- (d)}$$

from eqⁿ (b) — $\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}_d$

$$\Rightarrow \boxed{\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}} \quad \text{--- Maxwell eqⁿ 4}$$

∴ by (d)

— PHYSICAL SIGNIFICANCE — magnetic field can be generated by current density vector & time variation.

EQUATION OF CONTINUITY —

— Let q charges in a cube of volume V , come out from a surface. If there is flow of charge, that means current will be there.

— If current flows, there will be a decrease in charges present in the cube w.r.t. time. Hence,

$$I = - \frac{dq}{dt} \quad \text{--- (1)}$$

$$\text{where, } q = \int_V \rho \, dV$$

$$\text{Also, } I = \oint_S \vec{J} \cdot d\vec{S} \quad \left\{ \begin{array}{l} \vec{J} \rightarrow \text{current density} \\ \text{conduction} \end{array} \right. \quad \text{--- (2)}$$

from law of conservation of charge, equating (1) & (2)

$$\Rightarrow \oint_S \vec{J} \cdot d\vec{S} = - \frac{dq}{dt}$$

$$\Rightarrow \oint_S \vec{J} \cdot d\vec{S} = - \frac{d}{dt} \int_V \rho \, dV$$

Applying Gauss' Divergence theorem,

$$\Rightarrow \int_V (\vec{\nabla} \cdot \vec{J}) \, dV = - \frac{d}{dt} \int_V \rho \, dV$$

$$\Rightarrow \boxed{\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}} \longrightarrow \text{eqn of continuity}$$

- PHYSICAL SIGNIFICANCE - divergence of current density is always equal to negative rate of change of volume charge density.
- represents law of conservation of charge

POYNTING THEOREM — (Energy Conservation Law in Electromagnetism)

- According to Poynting theorem, power transformed into EM field is the sum of time rate change of EM energy, within certain volume & time rate of energy flowing out through boundary surface.

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} = - \int_V (\vec{E} \cdot \vec{J}) dV - \frac{\partial}{\partial t} \int_V \left(\frac{\epsilon E^2}{2} + \frac{\mu H^2}{2} \right) dV$$

↓
↓
↓

Rate of energy flow per unit area
Power loss
Rate of decrease of stored energy

* Poynting Vector - rate of energy transport per unit area.

- also known as instantaneous energy flux - density,

- represented by $\vec{S}$ or $\vec{P}$

$$\Rightarrow \vec{S} = \vec{E} \times \vec{H}$$

- $\vec{S}$ is $\perp$ to both $\vec{E}$ & $\vec{H}$.

- Unit $\rightarrow W/m^2$

- Given by JH Poynting in 1884.

UNIT-2SHM & EMT (FORMULAE LIST)SHM →

① $\vec{F}_{res} = -K\vec{x}$ (HOOKE'S LAW)
 → spring constant

(where, $x \rightarrow$ displacement, $\vec{F}_{res} \rightarrow$ restoring force)

② $\frac{K}{m} = \omega^2$ (where, $\omega \rightarrow$ angular frequency $= 2\pi f$)

③ $KE = \frac{m}{2} (A^2 \omega^2 \cos^2 \omega t)$

$PE = \frac{m}{2} (A^2 \omega^2 \sin^2 \omega t)$ (where, $m \rightarrow$ mass
 $A \rightarrow$ amplitude)

$TE = KE + PE = \frac{m}{2} A^2 \omega^2$

④ $v = \omega \sqrt{A^2 - x^2}$

$v_{max} = \omega A$

(where, $v \rightarrow$ velocity
 $v_{max} \rightarrow$ max^m velocity)

⑤ Time period,

$T = 2\pi \sqrt{\frac{m}{K}}$

⑥ frequency,

$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{K}{m}}$

⑦ $F = -\omega^2 x$ → standard eqⁿ of SHM

$F_{max} = \omega^2 A$

$F_{min} = 0$ (at mean position)

(where, $F \rightarrow$ force
 $F_{max} \rightarrow$ max^m force ; $F_{min} \rightarrow$ min^m force)

$$\mu_0 = 4\pi \times 10^{-7}$$

$$\epsilon_0 = 8.85 \times 10^{-12}$$

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EMT ↓

① Poynting theorem → $P = |\vec{E} \times \vec{H}| = EH \sin 90^\circ$
 { $\sin 90^\circ$: maximum transmission }

② Poynting vector → $P = \frac{\text{power transformed}}{\text{Area}}$
 { NOTE → for point source
 (area of sphere used) = $\frac{\text{point source}}{4\pi R^2}$ }

③ Relation b/w E & H → $\frac{E}{H} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.7 \Omega$
 (wave impedance)

④ Amplitude of electric field & Magnetic field vectors -
 $E_0 = E\sqrt{2}$; $H_0 = H\sqrt{2}$

⑤ Velocity of light → $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$
 (free space)

⑥ Velocity of EM wave through non-conducting medium → $v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$

⑦ Attenuation constant, α -

for Non-conducting medium → $\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$

for conducting medium → $\alpha = \sqrt{\frac{\mu \omega \sigma}{2}}$

(where, σ → surface charge density)

③ Skin depth, $\boxed{\delta / \delta = \frac{1}{\alpha}} \mu.m$